BACHELOR OF TECHNOLOGY

Computer Science and Engineering (AI & ML)

COURSE STRUCTURE

(Batches admitted from the Academic Year 2022 -2023)



MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

Accredited by NAAC with 'A+' Grade, UGC, Govt. of India

Affiliated to JNTUH, Approved by AICTE, ISO 9001:2015 Certified Institution

Maisammaguda, Dhullapally, Secunderabad, Kompally-500100

III Year B. Tech – I Semester (V Semester)

S. No	Course code	Subject	L	T	P	C	Max. Marks		Page No
							INT	EXT	
1	2266PC02	Machine Learning	3	0	0	3	40	60	92
2	2205PC10	Computer Networks	3	0	0	3	40	60	94
3		**Professional Elective-1**	3	0	0	3	40	60	
	2266PE01	Artificial Neural Networks							123
	2212PE02	Data Warehousing & Data Mining							125
	2205PE01	**Software Engineering**							127
	2262PE16	Information Security							129
	2205PE02	Computer Graphics							131
	2212PE01	Mobile Computing							133
4		**Professional Elective – II**	3	0	0	3	40	60	
	2266PE02	AI Search Method for Problem Solving							135
	2212PE04	Distributed Databases							137
	2205PE03	Software Testing Methodology							139
	2262PE017	Digital Forensics							141
	2212PE09	Multimedia							143
	2205PE06	Cloud Computing							145
5		**Open Elective-1**	3	0	0	3	40	60	
6	2200HS04	Design Thinking	2	0	0	1	40	60	96
7	2266PC62	Machine Learning Lab	0	0	3	1.5	40	60	98
8	2205PC66	Computer Networks Lab	0	0	3	1.5	40	60	100
9	2266PR03	Innovative Product Development -III	0	0	2	1	40	60	-
10	2200MC05	Technical Communications and Soft Skills*	2	0	0	0	100	0	101
		TOTAL	**17**	**1**	**8**	**20**	**460**	**540**	

*Mandatory course: Non-credit course, 50% of scoring is required for the award of the degree

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2200HS02) PROFESSIONAL ENGLISH

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To enrich students to express themselves appropriately and fluently in professional contexts.
- To enhance their employability through regular participation in group discussions and interview skills.
- To lay foundation with writing strategies for the future work place needs.
- To acquaint students with different components of professional presentation skills.
- To equip students with necessary training in listening to comprehend dialects of English language.

Course Outcomes:

- Draft coherent and unified paragraphs with adequate supporting details.
- Communicate effectively in English on issues and ideas with a reasonable degree of fluency and accuracy in different social settings and different kinds of social encounters.
- Demonstrate problem solving skills, decision-making skills, analytical skills.
- Comprehend and apply the pre-interview preparation techniques for successful interview.
- Achieve expertise in writing resume and cover letter formats.
- Understand the steps of writing 'Reports and Abstract'.

UNIT - I:

FOCUS ON LANGUAGE: Parts of speech - nominal compounds, noun phrases - relative pronoun - adjective - numerical, comparison and contrast, collocation and word combinations - verb – preposition and relative - conjunction- connectives, expressions of purpose and function, cause and effect - articles - adjectives - sentence pattern - tenses - voice - rewriting the sentences in impersonal/abbreviated passive grammatical structures - concord - sentence level verb noun agreement - gerund - rewriting infinitive into gerund - imperative - rewriting imperative into recommendation using should - word formation - varied grammatical function of the same word - affixes – prefix and suffix, number prefix, negative prefix - reported speech – editing strategies - conditional structures - real, unreal, no possibility, zero condition. writing formal definition - abbreviation and acronym - idioms and phrases, varieties of English - British versus American.

UNIT - II:

LISTENING SKILLS: Comprehension practice - vocabulary development - familiarity to varied types of spoken English and accents - developing ability to understand audio and video media - aiming at overcoming barriers to listening - listening to documentaries, radio news broadcasts, TV ews telecasts - active listening in discussions and to lectures - taking notes while listening - extracting information from listening.

UNIT - III:

SPEAKING SKILLS: Oral practice - role play - interplay - seminar - transcoding visual into oral - participating in short and longer conversation - voice record, replay, correction of intonation, pronunciation and flow of speech - phonemes - vowels, consonants, stress, rhythm, intonation – group discussion - participative learning - acquiring proficiency, fluency, accuracy in oral communication - speaking practice - developing confidence - extempore speech – learning professional/conversational etiquette – Oral presentation skills.

UNIT - IV:

READING SKILLS: Vocabulary extension - improving vocabulary - intensive reading - reading strategies - identifying topic sentence - guessing meaning from content - picking out specific information - professional reading - reading practice - predicting the content, critical and analytical reading - reading articles in English newspapers, sports magazines, encyclopedias - reading aloud, use of stress and intonation - reading and comprehending technical materials - cloze reading.

UNIT - V:

WRITING SKILLS: Discourse cohesion - improving writing skills, avoiding common grammatical errors in academic writing - extending the hints - writing shorter sentences - punctuation – dialogue writing - paragraph writing, problems and solutions, achieving coherence, transition words, sequence words - essays of descriptive and argumentative - writing instructions, use of imperatives - jumbled sentences into sequential paragraph using linguistic clues – report writing - technical reports, industry visit reports, events reports - writing recommendations - letter writing - formal and informal letters, e-mail writing - job application and resume, permission for in-plant training, business correspondence letters, calling for quotation, placing order, lodging complaint, persuasive letters - assignment writing - mini-project – telephonic etiquette- transcoding - transferring of information from text to pictorial/graphical representation and vice versa. Exercises apart from the text book shall also be referred for classroom tasks.

TEXT BOOKS:

1. Practical English Usage. Michael Swan. OUP, 1995.
2. Remedial English Grammar. F. T. Wood. Macmillan. 2007.
3. On Writing Well. William Zinsser. Harper ResourceBook.2001.

REFERENCES:

1. Study Writing. Liz Hamp-Lyons and Ben Heasly. Cambridge University Press, 2006.
2. Communication Skills. Sanjay Kumar and Pushpa Lata. Oxford University Press, 2011.
3. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2266PC02) MACHINE LEARNING

B.Tech. III Year I Sem **L T P C**

3 1 0 4

Course Objectives:

- To be able to formulate machine learning problems corresponding to different applications.
- To understand a range of machine learning algorithms along with their strengths and weaknesses.
- To understand the basic theory underlying machine learning.

Course Outcomes:

- Ideate various learning models (geometric, probabilistic, logic) and analyze key concepts like generalization, error, bias, and variance in learning processes.
- Apply linear and nonlinear classification techniques (e.g., logistic regression, perceptrons, and support vector machines) and assess model performance considering generalization and overfitting.
- Implement and evaluate distance-based models (e.g., k-means clustering, k-nearest neighbors, hierarchical clustering) and advanced techniques such as ensemble learning (e.g., bagging, boosting).
- Construct and analyze decision trees and rule-based models for classification, regression, and association rule mining, understanding their use in predictive and descriptive modeling.
- Develop algorithms for passive and active reinforcement learning (e.g., temporal-difference learning, policy search), and evaluate their application in real-world scenarios like game playing and robotics.
- Examine and apply the principles of model validation and regularization in various learning techniques to mitigate overfitting and enhance model reliability.

UNIT - I:

FOUNDATIONS OF LEARNING: Components of learning – learning models – geometric models – probabilistic models – logic models – grouping and grading – learning versus design – types of learning – supervised – unsupervised – reinforcement – theory of learning – feasibility of learning – error and noise– training versus testing – theory of generalization – generalization bound – approximation generalization trade off – bias and variance – learning curve.

UNIT - II:

INTRODUCTION TO TECHNOLOGY LANDSCAPE: Linear classification – univariate linear regression – multivariate linear regression – regularized regression – Logistic regression – perceptron's – multilayer neural networks – learning neural networks structures – support vector machines – soft margin SVM – going beyond linearity – generalization and overfitting – regularization – validation.

UNIT - III:

DISTANCE-BASED MODELS: Nearest neighbor models – K-means – clustering around medoids – silhouttes – hierarchical clustering – k-d trees – locality sensitive hashing – non-parametric regression – ensemble learning – bagging and random forests – boosting – meta learning

UNIT - IV:

TREE AND RULE MODELS: Decision trees – learning decision trees – ranking and probability estimation trees – regression trees – clustering trees – learning ordered rule lists – learning unordered rule lists– descriptive rule learning – association rule mining – first-order rule learning.

UNIT - V:

REINFORCEMENT LEARNING: Passive reinforcement learning – direct utility estimation – adaptive dynamic programming– temporal- difference learning – active reinforcement learning – exploration – learning an action utility function – Generalization in reinforcement learning – policy search – applications in game playing – applications in robot control.

TEXT BOOKS:

1. P. Flach, "Machine Learning: The art and science of algorithms that make sense of data", Cambridge University Press, 2012. (UNIT-I to IV)
2. M. Mohri, A. Rostamizadeh, and A. Talwalkar, "Foundations of Machine Learning", MIT Press, 2012. (UNITV)

REFERENCES:

1. Y. S. Abu-Mostafa, M. Magdon-Ismail, and H.-T. Lin, "Learning from Data", A ML Book Publishers, 2012.
2. K. P. Murphy, "Machine Learning: A probabilistic perspective", MIT Press, 2012.
3. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2007.
4. T. M. Mitchell, "Machine Learning", McGraw Hill, 1997.
5. S. Russel and P. Norvig, "Artificial Intelligence: A Modern Approach", Third Edition, Prentice Hall, 2009.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2205PC10) COMPUTER NETWORKS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To introduce the fundamental various types of computer networks.
- To demonstrate the TCP/IP and OSI models with merits and demerits.
- To explore the various layers of OSI Model.
- To introduce UDP and TCP Models.
- To introduce Encryption Security Mechanism.

Course Outcomes:

- Implement and demonstrate data link layer framing techniques, such as characterstuffing, bit stuffing, and character framing, to ensure data integrity in networkcommunication.
- Analyze error detection methods by implementing CRC-12, CRC-16, and CRC-32on a dataset and examining their effectiveness in identifying transmission errors.
- Implement and demonstrate the Stop-and-Wait protocol to illustrate a basic reliabledata transmission method in computer networks.
- Apply Dijkstra's algorithm to calculate the shortest path in a network graph,evaluating the efficiency of path computation for network routing.
- Construct routing tables for a subnet graph with given delays by implementingdistance vector and OSPF protocols, demonstrating routing table accuracy at eachnode.
- Implement DES and RSA algorithms to encrypt and decrypt text data, showcasingbasic cryptographic techniques for secure communication.

UNIT – I:

Data Communications: Components – Direction of Data flow – Networks – Components and Categories – Types of Connections – Topologies –Protocols and Standards – ISO / OSI model, Physical layer: Transmission modes, Multiplexing, Transmission Media, Switching, Circuit Switched Networks, Datagram Networks and Virtual Circuit Networks.

UNIT – II:

Data link layer: Introduction, Framing, and Error – Detection and Correction – Parity – LRC– CRC Hamming code, Flow and Error Control, Noiseless Channels, Noisy Channels, HDLC, Point to Point Protocols. 111 Medium Access sub layer: ALOHA, CSMA/CD, LAN– Ethernet IEEE 802.3, IEEE 802.5 – IEEE 802.11, Random access, Controlled access

UNIT – III:

Network layer: Logical Addressing, Internetworking, Tunneling, Address mapping, ICMP, IGMP, Forwarding, Uni-Cast Routing Protocols, Multicast Routing Protocols.

UNIT – IV:

Transport Layer: Process to Process Delivery, UDP and TCP protocols, Data Traffic, Congestion, Congestion Control, QoS, Integrated Services, Differentiated Services, QoS in Switched Networks.

UNIT – V:

Application Layer: Domain name space, DNS in internet, electronic mail, SMTP, FTP, WWW, HTTP, SNMP, Security – PGP – SSH.

TEXT BOOKS:

1. Data Communications and Networking, Behrouz A. Forouzan , Fourth Edition TMH, 2006.
2. Computer Networks, Andrew S Tanenbaum, 4th Edition. Pearson Education, PHI.

REFERENCES:

1. Data communications and Computer Networks, P.C .Gupta, PHI.
2. An Engineering Approach to Computer Networks, S. Keshav, 2nd Edition, Pearson Education.
3. Understanding communications and Networks, 3rd Edition, W.A. Shay, Cengage Learning.
4. Computer Networking: A Top-Down Approach Featuring the Internet. James F. Kurose & Keith W. Ross, 3 rd Edition, Pearson Education.
5. Data and Computer Communication, William Stallings, Sixth Edition, Pearson Education, 2000.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2200HS0) DESIGN THINKING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- Inculcate the fundamental concepts of design thinking
- Develop the students as a good designer by imparting creativity and problem solving ability
- Conceive, conceptualize, design and demonstrate innovative ideas using prototypes

Course Outcomes:

- Explain the principles of design thinking and its approaches
- Identify the empathy, define phases in human centered design problems.
- Develop an idea, build a prototype and test in design thinking context.
- Implement design thinking techniques for product innovation
- Provide a thinking space for innovation of new ventures, value propositions, new products or services.
- Use design thinking in business process models.

UNIT - I:

INTRODUCTION TO DESIGN THINKING:An insight into Design, origin of Design thinking, Design thinking Vs Engineering thinking, importance of Design thinking, Design Vs Design thinking, understanding Design thinking and its process models, application of Design thinking

UNIT - II:

EMPATHIZE IN DESIGN THINKING: Human-Centered Design (HCD) process - Empathize, Define, Ideate, Prototype and Test and Iterate. Role of Empathy in design thinking, methods and tools of empathy, understanding empathy tools. Explore define phase state users' needs and problems using empathy methods

UNIT - III:

IDEATION, PROTOTYPING AND TESTING: Ideation methods, brain storming, advantages of brain storming, methods and tools of ideations, prototyping and methods of prototyping, user testing methods, Advantages and disadvantages of user Testing/ Validation

UNIT - IV:

PRODUCT INNOVATION: Design thinking for strategic innovation, Definition of innovation, art of innovation, teams for innovation, materials and innovation in materials, definition of product and its classification. Innovation towards product design Case studies

UNIT - V:

DESIGN THINKING IN BUSINESS PROCESSES: Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs.

TEXT BOOKS:

1. Change by design, Tim Brown, 2009, Harper Collins

2. Engineering design, George E Dieter,4th Revised edition, 2009 McGraw Hill.

REFERENCES:

1. Design Thinking for Strategic Innovation, Idris Mootee,2013, John Wiley & Sons

2. Design Thinking-The Guidebook – Facilitated by the Royal Civil service Commission, Bhutan

3. Design Methods: A Structured Approach for Driving Innovation in Your Organization, Vijay Kumar, First Edition, 2012, Wiley

4. Human-Centered Design Toolkit: An Open-Source Toolkit to Inspire New Solutions in the Developing World, IDEO, Second Edition, 2011, IDEO

e-Resources & other digital material

1. https://www.interactiondesiqn.ora/literature/topics/desiqn-thinkinq

2. https://www.interactiondesiqn.prq/literature/article/how-tq-<eve'op-anempath\capproach-in-design-thinking

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2266PC62) MACHINE LEARNING LAB

B.Tech. III Year I Sem **L T P C**

0 0 3 1.5

Course Objective:

- The objective of this lab is to get an overview of the various machine learning
- This lab is to learn the machine learning techniques and can able to demonstrate those using python.

Course Outcomes:

- Apply Bayes' theorem to calculate conditional probabilities, implementing probabilistic reasoning in Python to solve classification and decision-making tasks.
- Extract and analyze structured data from databases using Python, preparing datasets for machine learning applications to enable effective data-driven insights.
- Develop predictive models using k-nearest neighbors (KNN) and k-means clustering, and assess the classification of new data points based on proximity and clustering outcomes.
- Evaluate the performance of different machine learning algorithms, including linear regression and Naïve Bayes, in predicting outcomes and classifying data, demonstrating an understanding of model accuracy and bias.
- Design and implement a genetic algorithm, showcasing optimization techniques by exploring solutions to complex problems through simulated evolution and natural selection.
- Construct and train a neural network for text classification using a back-propagation algorithm, effectively categorizing finite words and improving classification accuracy through iterative learning.

List of Experiments:

1. The probability that it is Friday and that a student is absent is 3 %. Since there are 5 school days in a week, the probability that it is Friday is 20 %. What is the probability that a student is absent given that today is Friday? Apply Baye's rule in python to get the result. (Ans:15%)
2. Extract the data from database using python.

3. Implement k-nearest neighbours classification using python

4. Given the following data, which specify classifications for nine combinations of VAR1 and VAR2 predict a classification for a case where VAR1=0.906 and VAR2=0.606, using the result of k-means clustering with 3 means (i.e., 3centroids)

VAR1	VAR2	CLASS
1.713	1.586	0
0.180	1.786	1
0.353	1.240	1
0.940	1.566	0
1.486	0.759	1

1.266	1.106	0
1.540	0.419	1
0.459	1.799	1
0.773	0.186	1

5. The following training examples map descriptions of individuals into high, medium and low.
credit- worthiness medium skiing design single twenties no ->high Risk
high golf trading married forties yes ->low Risk
low speedway transport married thirties yes ->med Risk
medium football banking single thirties yes ->low Risk
high flying media married fifties yes ->high Risk
low football security single twenties no ->med Risk
medium golf media single thirties yes ->med Risk
medium golf transport married forties yes ->low Risk
high skiing banking single thirties yes ->high Risk
low golf unemployed married forties yes ->high Risk
Input attributes are (from left to right) income, recreation, job, status, age-group, homeowner. Find the unconditional probability of `golf' and the conditional probability of `single' given `med Risk' in the dataset?

6. Implement linear regression using python.

7. Implement Naïve Bayes theorem to classify the English text

8. Implement an algorithm to demonstrate the significance of genetic algorithm

9. Implement the finite words classification system using Back-propagation algorithm

TEXT BOOKS:

1. Machine Learning – Tom M. Mitchell, MGH.

2. Fundamentals of Speech Recognition By Lawrence Rabiner and Biing – Hwang Juang.

REFERENCES:

1. Machine Learning: An Algorithmic Perspective, Stephen Marsland, Taylor & Francis.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2205PC66) COMPUTER NETWORKS LAB

B.Tech. III Year I Sem **L T P C**

0 0 3 1.5

Course Objectives:

- To understand the functionalities of various layers of OSI model.
- To understand the operating System functionalities.

Course Outcomes:

- Implement and demonstrate data link layer framing techniques, such as characterstuffing, bit stuffing, and character framing, to ensure data integrity in networkcommunication.
- Analyze error detection methods by implementing CRC-12, CRC-16, and CRC-32on a dataset and examining their effectiveness in identifying transmission errors.
- Implement and demonstrate the Stop-and-Wait protocol to illustrate a basic reliabledata transmission method in computer networks.
- Apply Dijkstra's algorithm to calculate the shortest path in a network graph,evaluating the efficiency of path computation for network routing.
- Construct routing tables for a subnet graph with given delays by implementingdistance vector and OSPF protocols, demonstrating routing table accuracy at eachnode.
- Implement DES and RSA algorithms to encrypt and decrypt text data, showcasingbasic cryptographic techniques for secure communication.

List of Experiments:

1. Implement the data link layer framing methods such as character, character stuffing, and bit stuffing.
2. Implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC.
3. Implement Stop and wait protocol.
4. Implement Dijkstra's algorithm to compute the Shortest path through a graph.
5. Take an example subnet graph with weights indicating delay between nodes. Now obtain Routing table art each node using distance vector routing algorithm
6. To implement Open Shortest Path First (OSPF) Routing Protocol
7. Take a 64 bit playing text and encrypt the same using DES algorithm
8. Using RSA algorithm encrypts a text data and Decrypt the same.

REFERENCES:

1. Data Communications and Networking – Behrouz A. Forouzan, 4th Edition McGraw Hill Education, 2006.
2. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose, K. W. Ross, 3rd Edition, Pearson Education.
3. Data communication and Networks – Bhusan Trivedi, Oxford University Press 2016.
4. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(2200MC05) TECHNICAL COMMUNICATIONS AND SOFT SKILLS

B.Tech. III Year I Sem **L T P C**

2 0 0 0

Course Objectives:

- To make the students recognize the role of Technical English in their academic and professional fields.
- To improve language proficiency and develop the required professional skills.
- To equip students with tools to organize, comprehend, draft short and long forms of technical work.

Course Outcomes:

- The students will be able to understand information which assists in completion of the assigned job tasks more successfully.
- Students will be able to communicate their ideas by writing projects, reports, instructions, diagrams and many other forms of professional writing
- Students will also be able to adhere to ethical norms of scientific communication
- Students will be able to strengthen their individual and collaborative work strategies
- Understanding the social issues and applying analytical abilities to solve the issues
- Students will be able to strengthen their individual and collaborative work strategies

UNIT - I:

Personal Evaluation: Self-Assessment and Self- Awareness - Self-Esteem - Perception and Attitudes – Values and Beliefs - Time Management- Concord.

UNIT - II:

Professional Communication: Extempore - Oral Presentations – Presentation Aids- Email Writing, Business Letter Writing - Memo Writing - Transformation of Sentences

UNIT – III:

Career Planning: Group Discussion, Interviews - Leadership Skills & Team Building - Personal Goal Setting and Career Planning - Complex Problem Solving - Creativity - Role and Responsibilities of an Engineer – Tenses.

UNIT - IV:

Technical Writing: Principles of Effective Writing - Editing Strategies to Achieve Appropriate Technical Style – Technical Report Writing – Voice.

UNIT – V:

Ethics and Responsibilities: Personality Development in Social and Office Settings – Netiquettes - Work Culture and Cubicle Etiquettes - Correction of Sentences.

PROFESSIONAL ELECTIVES

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2266PE01) ARTIFICIAL NEURAL NETWORKS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To understand Fuzzy set and logic control.
- To analyze Adaptive Fuzzy Systems.
- To understand the operation of Artificial Neural Networks.
- To understand mapping and recurrent networks operation.
- To observe various case studies.

Course Outcomes:

- Understand the structure and functioning of Artificial Neural Networks (ANNs), including biological and artificial neurons and activation functions.
- Apply mathematical foundations and learning mechanisms such as error-correction, Hebbian, and memory-based learning in ANN design.
- Implement and evaluate single-layer perceptrons for pattern classification and understand their limitations.
- Design and train multi-layer feedforward neural networks using backpropagation for functional approximation.
- Understand and apply Radial Basis Function (RBF) networks for pattern separability, interpolation, and approximation.
- Design, implement, and evaluate neural network models like perceptrons, multi-layer feedforward networks, and RBF networks for various pattern classification and approximation tasks.

UNIT – I:

Fuzzy Set Theory and Fuzzy Logic Control: Basic concepts of fuzzy sets- Operations on fuzzy sets- Fuzzy relation equations- Fuzzy logic control Fuzzification –Defuzzificatiuon- Knowledge base- Decision making logic Membership functions – Rule base.

UNIT – II:

Adaptive Fuzzy Systems: Performance index- Modification of rule base0- Modification of membership functions- Simultaneous modification of rule base and membership functions- Genetic algorithms Adaptive fuzzy system Neuro fuzzy systems.

UNIT – III:

Artificial Neural Networks: Introduction- History of neural networks- multilayer perceptions- Back propagation algorithm and its Variants- Different types of learning, examples.

UNIT – IV:

Mapping and Recurrent Networks: Counter propagation –Self organization Map- Congnitron and Neocognitron- Hopfield Net- Kohonnen Nets- Grossberg Nets- Art-I, Art-II reinforcement learning .

UNIT – V:

Case Studies: Application of fuzzy logic and neural networks to Measurement- Control- Adaptive Neural Controllers – Signal Processing and Image Processing.

TEXT BOOK:

Vallum B.R And Hayagriva V.R C++, Neural networks and Fuzzy logic, BPB Publications, New Delhi, 1996.

REFERENCE BOOKS:

1. Fuzzy logic & Neural Networks/ Chennakesava R. Alavala/ New Age International, 2008.
2. Neural Networks for control, Millon W. T, Sutton R.S and Werbos P. J, MIT Press 1992.
3. Fuzzy sets Fuzzy logic, Klir, G. J anfd Yuan B.B Prentice Hall oif India Pvt. Ltd.,, New Delhi.
4. Neural Networks and Fuzzy systems, Kosko.. Prentice hall of India Pvt. Ltd.,, New Delhi 1994.
5. Introduction to Fuzzy control, Dirankov D. Hellendoorn H, Reinfrank M., Narosa Publications House, New Delhi 1996.
6. Introduction to Artificial Neural systems, Zurada J. M Jaico Publishing House, New Delhi 1994.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2212PE02) DATA WAREHOUSING & DATA MINING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- Study data warehouse principles and its working learn data mining concepts.
- To understand association rules mining.
- Discuss classification algorithms.
- Learn how data is grouped using clustering techniques.

Course Outcomes:

- Student should be able to understand why the data warehouse in addition to database systems.
- Ability to perform the pre-processing of data and apply mining techniques on it.
- Ability to identify the association rules, classification and clusters in large datasets.
- Ability to solve real world problems in business and scientific information using data mining.
- Describe different methodologies used in data mining and data ware housing.
- Compare different approaches of data ware housing and data mining with various technologies.

UNIT – I:

Data Warehouse: Introduction to Data warehouse, Difference between operational database systems and data warehouses, Data warehouse Characteristics, Data warehouse Architecture and its Components, Extraction- Transformation-Loading, Logical(Multi- Dimensional), Data Modeling, Schema Design, Star and Snow-Flake Schema, Fact Constellation, Fact Table, Fully Addictive, Semi-Addictive, Non Addictive Measures; Fact- Less-Facts, Dimension Table Characteristics; OLAP Cube, OLAP Operations, OLAP Server Architecture- ROLAP, MOLAP and HOLAP.

UNIT – II:

Introduction: Fundamentals of data mining, Data Mining Functionalities, Classification of Data Mining systems, Data Mining Task Primitives, Integration of a Data Mining System with a Database or Data Warehouse System, Major issues in Data Mining.

Data Preprocessing: Need for Preprocessing the Data, Data Cleaning, Data Integration Transformation, Data Reduction, Discretization and Concept Hierarchy Generation.

UNIT - III:

Association Rules: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIOIRI Algorithm, The Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.

UNIT – IV:

Classification: Problem Definition, General Approaches to solving a classification problem, Evaluation of Classifiers , Classification techniques, Decision Trees- Decision tree Construction, Methods for Expressing attribute test conditions, Measures for Selecting the Best Split, Algorithm for Decision tree Induction; Naive- Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification- Algorithm and Characteristics. Prediction: Accuracy and Error measures. Evaluating the accuracy of a Classifier or a Predictor, Ensemble Methods.

UNIT - V:

Cluster Analysis: Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical Methods, Density-Based Methods, Grid-Based Methods, and Model based Clustering Methods, Outlier Analysis.

TEXT BOOKS:

1. Data Mining- Concepts and -1.chniques- Jiawei Han, Micheline Kamber, Morgan Kaufmann Publishers, Elsevier, 2Edition,2006.
2. Introduction to Data Mining, Psng-Ning Tan, Vipin Kumar, Michael Steinbanch, Pearson Educator.

REFERENCES:

1. Data Mining Techniques, Arun KPujari, 3rd Edition, Universities Press.
2. Data Warehousing Fundament's, Pualraj Ponnaiah, Wiley Student Edition.
3. The Data Warehouse Life CycleToolkit — Ralph Kimball, Wiley Student Edition.
4. Data Mining, Vikaram Pudi, PRddha Krishna, Oxford University Press.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2205PE01) SOFTWARE ENGINEERING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To comprehend the various software process models.
- To understand the types of software requirements and SRS document.
- To know the different software design and architectural styles.
- To learn the software testing approaches and metrics used in software development.
- To know about quality control and risk management.

Course Outcomes:

- To compare and select a process model for a business system.
- To identify and specify the requirements for the development of an application.
- To develop and maintain efficient, reliable and cost-effective software solutions.
- To critically think and evaluate assumptions and arguments of the client.
- Demonstrate an understanding of and apply current theories, models, and techniques that provide a basis for the software lifecycle
- Demonstrate an ability to use the techniques and tools necessary for engineering practice

UNIT - I:

Introduction to Software Engineering: The evolving role of software, Changing Nature of Software, Software myths. A Generic view of process: A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models. Process models: The waterfall model, Incremental process models, Evolutionary process models, The Unified process, Agility and Agile Process models of Agile Development and Tools.

UNIT - II:

Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document (IEEE FORMAT) and its contents. Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management. System models: Context Models, Behavioral models, Data models, Object models, structured methods. UML Diagrams.

UNIT - III:

Design Engineering: Design process and Design quality, Design concepts, the design model. Creating an architectural design: Software architecture, Data design, Architectural styles and patterns, Architectural Design. Object-Oriented Design: Objects and object classes, An Object-Oriented design process, Design evolution. Performing User interface design: Golden rules, User interface analysis and design, interface analysis, interface design steps.

UNIT -IV:

Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging. Product metrics: Software Quality, Metrics for Analysis Model, Metrics for Design Model, Metrics for source code, Metrics for testing, Metrics for maintenance. Metrics for Process and Products: Software Measurement, Metrics for software quality.

UNIT - V:

Risk management: Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan. Quality Management: Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

TEXT BOOKS:

1. Software Engineering A Practitioner's Approach, Roger S Pressman, 6th edition. McGraw Hill International Edition.
2. Software Engineering, Ian Sommerville, 7th edition, Pearson education.

REFERENCE BOOKS:

1. Software Engineering, A Precise Approach, Pankaj Jalote,Wiley India,2010.
2. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw- Hill,2008
3. Software Engineering, Principles and Practices, Deepak Jain, Oxford UniversityPress.
4. SoftwareEngineering1: Abstraction and modeling, Diner Bjorner, Springer Internationaledition,2006.
5. SoftwareEngineering2: Specification of systems and languages, Diner Bjorner, Springer Internationaledition2006.
6. Software Engineering Principles and Practice, Hans Van Vliet, 3rd edition, John Wiley &Sons Ltd.
7. Software Engineering3: Domains, Requirements, and Software Design, D. Bjorner, Springer International Edition.
8. Introduction to Software Engineering, R. J. Leach, CRC Press.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2262PE016) INFORMATION SECURITY

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- Understand various cryptographic algorithms.
- Understand the basic categories of threats to computers and networks
- Describe public-key cryptosystem.
- Describe the enhancements made to IPv4 by IPSec
- Understand Intrusions and intrusion detection
- Discuss the fundamental ideas of public-key cryptography.
- Generate and distribute a PGP key pair and use the PGP package to send an encrypted e- mail message.
- Discuss Web security and Firewalls

Course Outcomes:

- Evaluate various security attacks (interruption, interception, modification, and fabrication) and their implications on data integrity and confidentiality, demonstrating the ability to assess risk in different scenarios.
- Design and apply appropriate security services (confidentiality, authentication, integrity, non-repudiation, access control, and availability) to create a robust security model for internetwork security.
- Ideate classical and modern encryption techniques, including DES and Blowfish, evaluating their strengths and weaknesses in different contexts and demonstrating knowledge of cryptanalysis methods such as differential and linear cryptanalysis.
- Formulate and implement effective key management strategies, including public key cryptography principles (RSA, Diffie-Hellman, Elliptic Curve Cryptography), to ensure secure communication and data integrity.
- Assess the effectiveness of various digital signature standards and authentication protocols (e.g., Kerberos, X.509, PGP, and S/MIME) in providing secure and trustworthy communication.
- Implement and propose comprehensive security architectures that incorporate IP security (IPSec), web security protocols (SSL/TLS), and firewall design principles, addressing potential threats from intruders, viruses, and worms.

UNIT – I:

Security Concepts: Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.

UNIT – II:

Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm.

UNIT – III:

Cryptographic Hash Functions: Message Authentication, Secure Hash Algorithm (SHA-512), Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme. Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure

UNIT – IV:

Transport-level Security: Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH)
Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security

UNIT – V:

E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange Case Studies on Cryptography and security: Secure Multiparty Calculation, Virtual Elections, Single sign On, Secure Inter-branch Payment Transactions, Cross site Scripting Vulnerability.

TEXT BOOKS:

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition
2. Cryptography and Network Security: Atul Kahate, Mc Graw Hill, 3rd Edition.

REFERENCE BOOKS:

1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, Mc Graw Hill, 3rd Edition.
3. Information Security, Principles, and Practice: Mark Stamp, Wiley India.
4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH.
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2205PE02) COMPUTER GRAPHICS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To provide an introduction of fundamental concepts and theory of computer graphics.
- Topics covered include graphics systems and input devices; geometric representations and 2D/3D transformations; viewing and projections; illumination and color models; animation; rendering and implementation; visible surface detection.

Course Outcomes:

- Understand illumination and color models, including light sources, basic illumination models, and various color models like RGB, YIQ, CMY, HSV, and HLS.
- Apply two-dimensional geometric transformations, matrix representations, and homogeneous coordinates, and implement 2D viewing, window-to-viewport transformations, and clipping algorithms.
- Explore three-dimensional graphics concepts, object representations, curved lines, spline surfaces (Bezier and B-Spline), and perform 3D transformations, projections, and clipping.
- Learn multimedia system design, compression and decompression techniques, multimedia databases, and I/O technologies for digital audio, video, images, and animation.
- Understand multimedia authoring, user interfaces, hypermedia messaging, and create integrated multimedia systems and messages for distributed multimedia environments.
- Design and implement multimedia and hypermedia systems, focusing on compression, storage, retrieval technologies, and the integration of multimedia messages using tools like Blender Graphics.

UNIT – I:

Introduction: Application areas of Computer Graphics, overview of graphics systems, video-display devices, raster-scan systems, random scan systems, graphics monitors and work stations and input devices Output primitives: Points and lines, line drawing algorithms (Bresenham's and DDA Algorithm), midpoint circle and ellipse algorithms Polygon Filling: Scan-line algorithm, boundary-fill and flood-fill algorithms.

UNIT – II:

2-D geometrical transforms: Translation, scaling, rotation, reflection and shear transformations, matrix representations and homogeneous coordinates, composite transforms, transformations between coordinate systems. 2-D viewing: The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions, Cohen-Sutherland algorithms, Sutherland –Hodgeman polygon clipping algorithm.

UNIT – III:

3-D object representation: Polygon surfaces, quadric surfaces, spline representation, Hermite curve, Bezier curve and B-Spline curves, Bezier and B-Spline surfaces. Basic illumination models, polygon rendering methods.

UNIT – IV:

3-D Geometric transformations: Translation, rotation, scaling, reflection and shear transformations, composite transformations.

Viewing: Viewing pipeline, viewing coordinates, view volume and general projection transforms and clipping.

UNIT – V:

Computer animation: Design of animation sequence, general computer animation functions, raster animation, computer animation languages, key frame systems, motion specifications Visible surface detection methods: Classification, back-face detection, depth-buffer, BSP-tree methods and area sub-division methods.

TEXT BOOKS:

1. "Computer Graphics C version", Donald Hearn and M. Pauline Baker, Pearson Education
2. "Computer Graphics Principles & practice", second edition in C, Foley, Van Dam, Feiner and Hughes, Pearson Education.
3. Computer Graphics, Steven Harrington, TMH.

REFERENCES:

1. Procedural elements for Computer Graphics, David F Rogers, Tata Mc Graw hill, 2nd edition.
2. Principles of Interactive Computer Graphics", Neuman and Sproul, TMH.
3. Principles of Computer Graphics, Shalini Govil, Pai, 2005, Springer.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - I

(2212PE01) MOBILE COMPUTING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To make the student understand the concept of mobile computing paradigm, its novel applications and limitations.
- To understand the typical mobile networking infrastructure through a popular GSM protocol
- To understand the issues and solutions of various layers of mobile networks, namely MAC layer, Network Layer & Transport Layer
- To understand the database issues in mobile environments & data delivery models.
- To understand the ad hoc networks and related concepts.
- To understand the platforms and protocols used in mobile environment.

Course Outcomes:

- Able to think and develop new mobile application.
- Able to take any new technical issue related to this new paradigm and come up with a solution(s).
- Able to develop new ad hoc network applications and/or algorithms/protocols.
- Able to understand & develop any existing or new protocol related to mobile environment.
- Analyze various protocols of all layers for mobile and ad hoc wireless communication networks.
- To understand IP and TCP layers of Mobile Communication.

UNIT - I:

WIRELESS COMMUNICATION: Introduction - Frequencies and Regulations - Signals - Antennas - Propagation Ranges and Effects – Multipath Propagation - Effects of Mobility - Multiplexing - Modulation and Shift Keying – Spread Spectrum - Frequency Hopping and Direct Sequence- Medium Access Control – Specialized MAC – SDMF-FDMA-TDMA-CDMA- Mobile Computing – Paradigm, Promises/Novel Applications and Impediments and Architecture; Mobile and Handheld Devices, Limitations of Mobile and Handheld Devices.

UNIT - II:

TELECOMMUNICATION SYSTEMS: GSM – Services, System Architecture, Radio Interfaces, Protocols, Localization, Calling, Handover, Security, New Data Services, GPRS, CSHSD, DECT- - UMTS and IMT- 2000- Architecture, User Equipment, RNS, UTRAN,

Node B, RNC functions - W- CDMA - HSPA+, HSUPA, HSDPA+ - Satellite systems – History- Applications- Basics- Routing- Localization-Handover-Examples.

UNIT - III:

Wireless LAN and Mobile Network Layer: Wireless LAN -Infrared vs radio transmission - Infrastructure and ad-hoc network 205- IEEE 802.11- HIPER LAN-Bluetooth Mobile Network Layer- Mobile IP- Dynamic host configuration protocol- Mobile ad-hoc networks

UNIT - IV:

Mobile Transport Layer & Database issues: Mobile Transport Layer: Conventional TCP/IP Protocols, Indirect TCP, Snooping TCP, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Data Base Issues: Database Hoarding & Caching Techniques, Client- Server Computing & Adaptation, Transactional Models, Query processing, Data Recovery Process & QoS Issues.

UNIT - V:

Mobile Application Development: File Systems- World wide web- Wireless application protocol (version 1.x)- i-mode- SyncML- WAP 2.0- Mobile Platform- Palm OS, Windows CE, Symbian OS, Linux for Mobile Devices, Android.

TEXT BOOKS:

1. Jochen Schiller, “Mobile Communications”, Addison-Wesley, Second Edition 2009.
2. Raj Kamal, “Mobile Computing”, Oxford University Press, 2007.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2266PE02) AI SEARCH METHOD FOR PROBLEM SOLVING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course objectives:

- To understand various searching techniques
- To understand various functional planning

Course outcomes:

- Understand the history of AI, key philosophical concepts, and the foundational ideas of state space search methods, including depth-first search, breadth-first search, and iterative deepening.
- Apply heuristic search techniques such as best-first search and hill climbing, and solve optimization problems like the Traveling Salesman Problem (TSP) using stochastic local search and population-based methods like genetic algorithms and ant colony optimization.
- Implement advanced search algorithms like A*, IDA*, and RBFS for optimal pathfinding, and understand game theory concepts for game playing using algorithms like Minimax and Alpha-Beta pruning.
- Learn problem decomposition techniques such as Means-Ends Analysis and explore rule-based expert systems, including production systems, inference engines, and various logical deduction methods like forward and backward chaining.
- Apply constraint satisfaction problem (CSP) techniques for problem-solving, including consistency-based diagnosis, backtracking algorithms, and forward checking methods for ensuring the validity of solutions.
- Integrate search algorithms, automated planning, and constraint processing methods to solve complex AI problems, such as optimizing game strategies, automated planning, and constraint satisfaction.

UNIT - I:

Introduction: History, Can Machines think?, Turing Test, Winograd Schema Challenge, Language and Thought, Wheels & Gears Introduction: Philosophy, Mind, Reasoning, Computation, Dartmouth Conference, The Chess Saga, Epiphenomena State Space Search: Depth First Search, Breadth First Search, Depth First Iterative Deepening.

UNIT - II:

Heuristic Search: Best First Search, Hill Climbing, Solution Space, TSP, Escaping Local Optima, Stochastic Local Search Population Based Methods: Genetic Algorithms, SAT, TSP, emergent Systems, Ant Colony Optimization Finding Optimal Paths: Branch & Bound, A*, Admissibility of A*, Informed Heuristic Functions.

UNIT - III:

Space Saving Versions of A*: Weighted A*, IDA*, RBFS, Monotone Condition, Sequence Alignment, DCFS, SMGS, Beam Stack Search Game Playing: Game Theory, Board Games and Game Trees, Algorithm Minimax, Alpha Beta and SSS* Automated Planning: Domain Independent Planning, Blocks World, Forward &Backward Search, Goal Stack Planning, Plan Space Planning.

UNIT – IV:

Problem Decomposition: Means Ends Analysis, Algorithm Graph plan, Algorithm AO* Rule Based Expert Systems: Production Systems, Inference Engine, Match-Resolve-Execute, Rete Net Deduction as Search: Logic, Soundness, Completeness, First Order Logic, Forward Chaining, Backward Chaining.

UNIT – V:

Constraint Processing: CSPs, Consistency Based Diagnosis, Algorithm Backtracking, Arc Consistency, Algorithm Forward Checking.

TEXT BOOKS:

1. Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013.

REFERENCES:

1. Stefan Edelkamp and Stefan Schroedl. Heuristic Search: Theory and Applications, Morgan Kaufmann, 2011.
2. John Haugeland, Artificial Intelligence: The Very Idea, A Bradford Book, The MIT Press, 1985.
3. Pamela McCorduck, Machines Who Think: A Personal Inquiry into the History and Prospects of Artificial Intelligence, A Peters/CRC Press; 2 edition, 2004.
4. Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill, 1991.
5. Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition, Prentice Hall, 2009.
6. Eugene Charniak, Drew McDermott. Introduction to Artificial Intelligence, Addison-Wesley, 1985.
7. Patrick Henry Winston. Artificial Intelligence, Addison-Wesley, 1992.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2212PE04) DISTRIBUTED DATABASES

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Outcomes:

- Evaluate the key architectural models and design strategies for Distributed Database Management Systems (DDBMS), including fragmentation and allocation, to recommend optimal design solutions for distributed environments.
- Analyze complex distributed query processing and decomposition techniques to optimize localization of distributed data and apply algorithms for efficient distributed query optimization.
- Formulate strategies for distributed transaction management by applying advanced concurrency control algorithms and deadlock management techniques to ensure transaction integrity and serializability.
- Assess the reliability and fault-tolerance mechanisms in DDBMS, including protocols for managing site failures and network partitioning, to enhance the resilience of distributed systems.
- Develop parallel database system architectures with a focus on data placement, query processing, and load balancing to improve the performance and scalability of distributed data processing.
- Design distributed object-oriented database solutions, applying object-oriented concepts such as inheritance, object identity, and persistence to compare and contrast the functionality of OODBMS and ORDBMS models.

UNIT – I:

Distributed Data Processing, Distributed Database System, Promises of DDBSs, Problem areas. Distributed DBMS Architecture: Architectural Models for Distributed DBMS, DDMBS Architecture. Distributed Database Design: Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.

UNIT - II:

Query processing and decomposition: Query processing objectives, characterization of query processors, layers of query processing, query decomposition, localization of distributed data. Distributed query Optimization: Query optimization, centralized query optimization, distributed query optimization algorithms.

UNIT - III:

Transaction Management: Definition, properties of transaction, types of transactions, distributed concurrency control: serializability, concurrency control mechanisms & algorithms, time - stamped & optimistic concurrency control Algorithms, deadlock Management.

UNIT - IV:

Distributed DBMS Reliability: Reliability concepts and measures, fault-tolerance in distributed systems, failures in Distributed DBMS, local & distributed reliability protocols, site failures and network partitioning. Parallel Database Systems: Parallel database system architectures, parallel data placement, parallel query processing, load balancing, database clusters.

UNIT - V:

Distributed object Database Management Systems: Fundamental object concepts and models, object distributed design, architectural issues, object management, distributed object storage, object query Processing. Object Oriented Data Model: Inheritance, object identity, persistent programming languages, persistence of objects, comparison OODBMS and ORDBMS.

TEXT BOOKS:

1. M. Tamer OZSU and Patuck Valduriez: Principles of Distributed Database Systems, Pearson Edn. Asia, 2001.
2. Stefano Ceri and Giuseppe Pelagatti: Distributed Databases, McGraw Hill.

REFERENCES:

1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom: “Database Systems: The Complete Book”, Second Edition, Pearson International Edition.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2205PE03) SOFTWARE TESTING METHODOLOGY

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To learn and understand the tools and technique for software testing and its practice in the industry.
- To be aware of the differences between the various testing strategies.
- To know the taxonomy and purpose of software testing tools.
- Ability to learn path testing, domain testing.
- To know the data flow testing.

Course Outcomes:

- Understand the purpose of software testing, various testing models, the consequences of bugs, and the taxonomy of bugs.
- Apply flow graph and path testing techniques, including path sensitization, path instrumentation, and transaction flow testing to ensure code quality.
- Implement data flow testing strategies and domain testing techniques to identify issues in program behavior and testing interfaces.
- Apply path products, path expressions, regular expressions, and logic-based testing techniques like decision tables and KV charts for comprehensive test case design.
- Understand and apply state graph concepts, transition testing, and testability tips to improve the reliability of state-driven systems.
- Use graph matrices, path expressions, and state graphs to design efficient test cases, and apply reduction algorithms and tools for enhanced test coverage.

UNIT - I:

Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs.

UNIT - II:

Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing. Transaction Flow Testing: Transaction flows, transaction flow testing techniques.

UNIT - III:

Dataflow testing: Basics of dataflow testing, strategies in dataflow testing, application of dataflow testing. Domain Testing:- domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.

UNIT - IV:

Paths, Path products and Regular expressions: Path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing : Overview, decision tables, path expressions, kv charts, specifications.

UNIT - V:

State, State Graphs and Transition testing: State graphs, good & bad state graphs, state testing, Testability tips. Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools.

TEXT BOOKS:

1. Software Testing techniques - Boris Beizer, Dreamtech, second edition.
2. Software Testing Tools – Dr. K.V.K.K. Prasad, Dreamtech.

REFERENCE BOOKS:

1. The craft of software testing - Brian Marick, Pearson Education.
2. Software Testing Techniques – SPD(Oreille).
3. Software Testing in the Real World – Edward Kit, Pearson.
4. Effective methods of Software Testing, Perry, John Wiley.
5. Art of Software Testing – Meyers, John Wiley.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2262PE017) DIGITAL FORENSICS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objective:

- To correctly define and cite appropriate instances for the application of computer forensics
- Correctly collect and analyze computer forensic evidence.
- Identify the essential and up-to-date concepts, algorithms, protocols, tools, and methodology of Computer Forensics.

Course Outcomes:

- Describe Forensic science and Digital Forensic concepts.
- Understand the basic terminology of cybercrimes.
- Apply a number of different computer forensic tools to a given scenario.
- Understand the basics of computer forensics.
- Analyze and validate digital evidence data.
- Analyze acquisition methods for digital evidence related to system security

UNIT - I:

Cyber Crime and computer crime Introduction to Digital Forensics, Definition and types of cybercrimes, electronic evidence and handling, electronic media, collection, searching and storage of electronic media, introduction to internet crimes, hacking and cracking, credit card and ATM frauds, web technology, cryptography, emerging digital crimes and modules.

UNIT - II:

Basics of Computer Computer organisation, components of computer- input and output devices, CPU, Memory hierarchy, types of memory, storage devices, system softwares, application softwares, basics of computer languages.

UNIT - III:

Computer Forensics Definition and Cardinal Rules, Data Acquisition and Authentication Process, Windows Systems-FAT12, FAT16, FAT32 and NTFS, UNIX file Systems, mac file systems, computer artifacts, Internet Artifacts, OS Artifacts and their forensic applications

UNIT - IV:

Forensic Tools and Processing of Electronic Evidence Introduction to Forensic Tools, Usage of Slack space, tools for Disk Imaging, Data Recovery, Vulnerability Assessment Tools, Encase and FTK tools,

UNIT - V:

Anti Forensics and probable counters, retrieving information, process of computer forensics and digital investigations, processing of digital evidence, digital images, damaged SIM and data recovery, multimedia evidence, retrieving deleted data: desktops, laptops and mobiles, retrieving data from slack space, renamed file, ghosting, compressed files.

TEXTBOOKS:

1. C. Altheide& H. Carvey Digital Forensics with Open Source Tools, Syngress, 2011.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2212PE09) MULTIMEDIA

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To introduce the basics of multimedia information storage technology, techniques for analysis, representation and retrieval that is commonly used in industry.
- To compare and contrast information retrieval models and internal mechanisms such as Boolean, Probability, and Vector Space Models.
- To outline the structure of queries and media elements.
- To critically evaluate Multimedia retrieval system effectiveness and improvement techniques

Course Outcomes:

- Learn the basics of multimedia information storage technology, techniques for analysis, representation and retrieval that is commonly used in industry.
- Compare and contrast information retrieval models and internal mechanisms such as Boolean, Probability, and Vector Space Models.
- Outline the structure of queries and media elements.
- Critically evaluate Multimedia retrieval system effectiveness and improvement techniques
- Design an image using image editing tools and apply effectively. Create animated sequence with titles applying the principles of animation.
- Apply acquired knowledge in the field of multimedia for the good cause like advertisement in practice and independently continue to expand knowledge in this field.

UNIT – I:

FUNDAMENTAL MEDIA UNDERSTANDING: Introduction – Media Types – Media Understanding – Description of Audio, Visual spectral and Video - Storage networks, storage medium.

UNIT – II:

TEXT RETRIEVAL AND MUSIC: Text Information retrieval: Information retrieval system-catalog and indexing – automatic indexing – term clustering – User search Techniques-Information Visualization- Fundamentals – Instantaneous Features - Intensity - Tonal Analysis - Musical Genre, Similarity and Mood.

UNIT - III:

IMAGE RETRIEVAL: content-based image retrieval; techniques; feature extraction; integration; similarity; feature in INDEXING; interactive retrieval; MPEG-7 standard

UNIT – IV:

VIDEO RETRIEVAL: Content Based Video Retrieval - Video Parsing – Video abstraction and Summarization– Video Content Representation, Indexing and retrieval –Video Browsing Schemes–Example of Video Retrieval Systems

UNIT – V:

RETRIEVAL METRICS AND MODERN IR: Average recall and average precision - Harmonic mean - Evaluation of a search engine – Relevance Issue – Kappa Measure – Quality versus Quantity, possible factors which influence outcome of a search – Grandfield Experimental Study. Introduction- parallel IR – Distributed IR – trends and research Issue.

REFERENCES:

1. Brusilovsky, Peter et.al. The Adaptive Web: Methods and Strategies of Web Personalization. Berlin: Springer, 2007.
2. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze," Introduction to Information Retrieval" , Cambridge University Press, 2008
3. Ricci, F.; Rokach, L.; Shapira, B.; Kantor, P.B. (Eds.), Recommender Systems Handbook. 1st Edition., 2011.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - II

(2205PE06) CLOUD COMPUTING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- This course provides an insight into cloud computing.
- Topics covered include- distributed system models
- To different cloud service models, service-oriented architectures, cloud programming and software environments, resource management.

Course Outcomes:

- Ability to understand various service delivery models of a cloud computing architecture.
- Ability to understand the ways in which the cloud can be programmed and deployed.
- Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing.
- Apply the fundamental concepts in datacenters to understand the tradeoffs in power, efficiency and cost.
- Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing.
- Analyze various cloud programming models and apply them to solve problems on the cloud.

UNIT – I:

Computing Paradigms: High-Performance Computing, Parallel Computing, Distributed Computing, Cluster Computing, Grid Computing, Cloud Computing, Bio computing, Mobile Computing, Quantum Computing, Optical Computing, Nano computing.

UNIT – II:

Cloud Computing Fundamentals: Motivation for Cloud Computing, The Need for Cloud Computing, Defining Cloud Computing, Definition of Cloud computing, Cloud Computting Is a Service, Cloud, Computing Is a Platform, Principles of Cloud computing, Five Essential Characteristics, Four Cloud Deployment Models.

UNIT – III:

Cloud Computing Architecture and Management: Cloud architecture, Layer, Anatomy of the Cloud, Network Connectivity in Cloud Computing, Applications, on the Cloud, Managing the Cloud, Managing the Cloud Infrastructure Managing the Cloud application, Migrating Application to Cloud, Phases of Cloud Migration Approaches for Cloud Migration.

UNIT – IV:

Cloud Service Models: Infrastructure as a Service, Characteristics of IaaS. Suitability of IaaS, Pros and Cons of IaaS, Summary of IaaS Providers, Platform as a Service. Characteristics of PaaS, Suitability of PaaS, Pros and Cons of PaaS, Summary of PaaS Providers, Software as a

Service, Characteristics of SaaS, Suitability of SaaS, Pros and Cons of SaaS, Summary of SaaS Providers, Other Cloud Service Models.

UNIT – V:

Cloud Service Providers: EMC, EMC IT, Captiva Cloud Toolkit, Google, Cloud Plaform, Cloud Storage, Google Cloud Connect, Google Cloud Print, Google App Engine, Amazon Web Services, Amazon Elastic Compute Cloud, Amazon Simple Storage Service, Amazon Simple Queue , service, Microsoft, Windows Azure, Microsoft Assessment and Planning Toolkit, SharePoint, IBM, Cloud Models, IBM Smart Cloud, SAP Labs, SAP HANA Cloud Platform, Virtualization Services Provided by SAP, Sales force, Sales Cloud, Service Cloud: Knowledge as a Service, Rack space, VMware, Manjra soft, Aneka Platform.

TEXT BOOKS:

1. Essentials of cloud Computing: K. Chandra sekhran, CRC press, 2014.

REFERENCES:

1. Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg and Andrzej M. Goscinski, Wiley, 2011.
2. Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.
3. Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Tim Mather, Subra Kumaraswamy, Shahed Latif, O'Reilly, SPD, rp2011.

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

PROFESSIONAL ELECTIVE - III

(2212PE03) INFORMATION RETRIEVAL SYSTEMS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Outcomes:

- Study fundamentals of DBMS, Data warehouse and Digital libraries
- Learn various preprocessing techniques and indexing approaches in text mining
- Know various clustering approaches and study different similarity measures
- Study various search techniques in information retrieval systems
- Know different cognitive approaches used in text retrieval systems and evaluation approaches
- Study retrieval in multimedia systems and know various evaluation measures
- Know about query languages and online IR system.

Course Outcomes:

- Understand the retrieval of relevant information from a text database.
- Recognize the Boolean Model, Vector Space Model, and Probabilistic Model.
- Understand retrieval utilities.
- Understand different formatting tags
- Understand cross-language information retrieval
- Understand the clustering techniques and determine the efficiency.

UNIT – I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses. Information Retrieval System Capabilities: Search, Browse, Miscellaneous.

UNIT – II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction. Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT – III:

Automatic Indexing: Classes of automatic indexing, Statistical indexing, Natural language, Concept indexing, Hypertext linkages Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT - IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean systems, Searching the Internet and hypertext. Information Visualization: Introduction, Cognition and perception, Information visualization technologies.

UNIT – V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems. Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXTBOOK:

1. Information Storage and Retrieval Systems: Theory and Implementation by Gerald J. Kowalski, Mark T. Maybury , Second Edition, Kluwer Academic Publishers.

REFERENCES:

1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms, Prentice Hall, 1992.
2. Modern Information Retrival By Yates Pearson Education.
3. Information Storage & Retieval By Robert Korfhage – John Wiley & Sons.

OPEN ELECTIVES

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

OPEN ELECTIVE - I

(2266OE01) KNOWLEDGE REPRESENTATION& REASONING

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objective:

- The main objective of the course is to provide the students with basic understanding of (logic-based) knowledge representation and reasoning.
- The course presents important knowledge representation languages, their use and limitations, and techniques and tools for reasoning over them.
- After completing the course, the students will be able to handle the main tools and dive deeper into the theoretical results of the area.

Course Outcomes:

- Understand the main knowledge representation languages and their limitations
- Learn to solve different reasoning tasks being aware of their complexity
- Develop methods to derive logical consequences from explicit knowledge
- Learn to analyse logical formulas and understand their intended meaning
- Apply logic-based knowledge representation to advanced AI applications

UNIT – I:

An Introduction to Knowledge Engineering, the history of knowledge-based expert systems, Types of Knowledge based systems, Characteristics of current expert systems, Basic concepts for building expert systems.

UNIT – II:

Knowledge Representation & Reasoning- Logic, Rules & representation, Developing Rules based system & Semantic Networks, Knowledge Acquisition, Knowledge representation schemes.

UNIT – III:

Building the Expert System, architecture of expert systems, Constructing an expert system, methods of inference. Expert systems shells, Development of environments, Use of AI Language.

UNIT – IV:

Conceptual data analysis; plausible reasoning techniques, Tools for building expert systems. Reasoning under uncertainty, Introduction to Clips, Pattern matching, Modular design and execution control.

UNIT – V:

Production-rule programming, Issues and case studies. Comparing different Approaches. Language and Tools for Knowledge Engineering, Expert system design examples, A Case Study in Knowledge Engineering.

TEXT BOOKS:

1. Joseph C Giarratano, Gary D Riley, Expert Systems – Principles & Programming, Third Edition, Course Technology Publishers.
2. Simon Kendal & Malcolm Creen, An Introduction to Knowledge Engineering, Springer Publishers, 2007.

REFERENCE BOOKS:

1. Buchanan, B. B. & Shortliffe, E. H. Building Expert Systems with Production Rules: The Mycin Experiments. Addison-Wesley Publishing Company
2. Davis, R. & Lenat, D. B. Knowledge-Based Systems in Artificial Intelligence. McGraw-Hill International Book Company
3. Hayes-Roth, F., Waterman, D. A. & Lenat, D. B. (eds) Building Expert Systems. Addison Wesley Publishing Company, Inc.
4. Torsun, I. S. Expert Systems: State of the Art, Addison-Wesley Publishing Company

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OPEN ELECTIVE - I

(2266OE02) NEURAL NETWORKS

B.Tech. III Year I Sem **L T P C**

3 0 0 3

Course Objectives:

- To understand Fuzzy set and logic control.
- To analyze Adaptive Fuzzy Systems.
- To understand the operation of Artificial Neural Networks.
- To understand mapping and recurrent networks operation.
- To observe various case studies.

Course Outcomes:

- Be able to analyze a problem for NN solution in terms of these methods.
- Have an awareness of the computational theory underlying NN.
- Have a working knowledge of a typical neural network simulation
- Experience in programming NN applications from scratch.
- Have knowledge of sufficient theoretical background to be able to reason about the behavior of neural networks.
- Have knowledge of sufficient theoretical background to be able to reason about the behavior of neural networks.

UNIT – I:

Fuzzy Set Theory and Fuzzy Logic Control: Basic concepts of fuzzy sets- Operations on fuzzy sets- Fuzzy relation equations- Fuzzy logic control Fuzzification –Defuzzificatiuon- Knowledge base- Decision making logic Membership functions – Rule base.

UNIT – II:

Adaptive Fuzzy Systems: Performance index- Modification of rule base0- Modification of membership functions- Simultaneous modification of rule base and membership functions- Genetic algorithms Adaptive fuzzy system Neuro fuzzy systems.

UNIT – III:

Artificial Neural Networks: Introduction- History of neural networks- multilayer perceptions- Back propagation algorithm and its Variants- Different types of learning, examples.

UNIT – IV:

Mapping and Recurrent Networks: Counter propagation –Self organization Map- Congnitron and Neocognitron- Hopfield Net- Kohonnen Nets- Grossberg Nets- Art-I, Art-II reinforcement learning .

UNIT – V:

Case Studies: Application of fuzzy logic and neural networks to Measurement- Control- Adaptive Neural Controllers – Signal Processing and Image Processing.

TEXT BOOK:

1. Vallum B.R And Hayagriva V.R C++, Neural networks and Fuzzy logic, BPB Publications, New Delhi, 1996.

REFERENCE BOOKS:

1. Fuzzy logic & Neural Networks/ Chennakesava R. Alavala/ New Age International, 2008.
2. Neural Networks for control, Millon W. T, Sutton R.S and Werbos P. J, MIT Press 1992.
3. Fuzzy sets Fuzzy logic, Klir, G. J anfd Yuan B.B Prentice Hall oif India Pvt. Ltd.,, New Delhi.
4. Neural Networks and Fuzzy systems, Kosko.. Prentice hall of India Pvt. Ltd.,, New Delhi 1994.
5. Introduction to Fuzzy control, Dirankov D. Hellendoorn H, Reinfrank M., Narosa Publications House, New Delhi 1996.
6. Introduction to Artificial Neural systems, Zurada J. M Jaico Publishing House, New Delhi 1994.